

Experiment 6: Apply K-Means and Hierarchical Clustering on sample datasets

Aim: Apply K-Means and Hierarchical Clustering on sample datasets.

Theory:

1. Dataset Source

Dataset Name: **Seeds Dataset**

Source: Kaggle

Link: <https://www.kaggle.com/datasets/rwzhang/seeds-dataset>

The dataset contains measurements of wheat seeds from three different varieties.

2. Dataset Description

The Seeds dataset consists of **210 samples** with **7 continuous numerical features**.

Features include:

- Area
- Perimeter
- Compactness
- Length of Kernel
- Width of Kernel
- Asymmetry Coefficient
- Length of Kernel Groove

Target Variable:

- Class (3 varieties of wheat seeds)

Since clustering is an **unsupervised learning technique**, the class label was removed during clustering.

Dataset Characteristics:

- All features are numerical

- No major missing values
- Suitable for distance-based clustering algorithms

Type of Problem:

Unsupervised Learning (Clustering)

3. Mathematical Formulation of the Algorithms

A) K-Means Clustering

K-Means partitions data into K clusters by minimizing within-cluster variance.

Distance formula (Euclidean Distance):

$$d(x, c) = \sqrt{\sum_{i=1}^n (x_i - c_i)^2}$$

Objective Function:

$$J = \sum_{k=1}^K \sum_{x_i \in C_k} ||x_i - \mu_k||^2$$

Where:

- C_k = Cluster k
- μ_k = Centroid of cluster k
- x_i = Data point

The goal is to minimize total intra-cluster distance.

B) Hierarchical Clustering (Agglomerative)

Agglomerative clustering starts with each data point as a separate cluster and merges clusters iteratively.

Ward's method minimizes variance between merged clusters.

Distance between clusters is calculated using linkage criteria.

Silhouette Score formula:

$$S = \frac{b - a}{\max(a, b)}$$

Where:

- a = Average intra-cluster distance
- b = Average nearest-cluster distance

Range:

- +1 → Strong clustering
- 0 → Overlapping clusters
- -1 → Poor clustering

4. Algorithm Limitations

K-Means Limitations

- Requires predefined number of clusters (K)
- Sensitive to outliers
- Works best for spherical clusters
- Results depend on centroid initialization

Hierarchical Clustering Limitations

- Computationally expensive for large datasets
- Cannot undo cluster merging
- Selecting optimal dendrogram cut is subjective

5. Methodology / Workflow

The experiment followed these steps:

1. Load Seeds dataset from Kaggle
2. Remove class column (unsupervised learning)

3. Apply StandardScaler for feature scaling
4. Apply K-Means clustering with $K = 3$
5. Apply Agglomerative Clustering with 3 clusters
6. Evaluate using Silhouette Score
7. Visualize clusters
8. Plot dendrogram for hierarchical clustering
9. Compare performance

Workflow Representation:

Dataset → Preprocessing → Scaling → Clustering → Evaluation → Visualization

6. Performance Analysis

K-Means Silhouette Score: **0.4038**

Hierarchical Silhouette Score: **0.3922**

Interpretation:

Both algorithms produced reasonably well-separated clusters.

K-Means achieved a slightly higher silhouette score, indicating better cluster compactness and separation.

Hierarchical clustering provided a dendrogram that visually explains cluster merging structure.

Thus, K-Means performed slightly better in terms of clustering quality.

7. Hyperparameter Tuning

For K-Means:

- Number of clusters (K) selected as 3 (based on dataset classes)
- Can be optimized using Elbow Method

For Hierarchical Clustering:

- Number of clusters set to 3
- Ward linkage method used

Hyperparameter tuning impacts cluster separation and silhouette score.

Code:

```
# Import Required Libraries
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
from sklearn.cluster import KMeans, AgglomerativeClustering
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score
from scipy.cluster.hierarchy import dendrogram, linkage
import os
```

```
# Load Dataset from KaggleHub
```

```
import kagglehub
```

```
# Download dataset
```

```
path = kagglehub.dataset_download("rwzhang/seeds-dataset")
```

```
print("Dataset Path:", path)
```

```
# Check files inside folder
```

```
print("Files:", os.listdir(path))
```

```
# Load CSV file (adjust name if needed)
```

```
df = pd.read_csv(path + "/seeds_dataset.csv")
```

```
print("Dataset Shape:", df.shape)
```

```
print(df.head())
```

```
# Prepare Data for Clustering
```

```
# Drop class column (unsupervised learning)
```

```
X = df.drop("class", axis=1)
```

```
# Standardize features
```

```
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)
```

```
# K-MEANS CLUSTERING
```

```
# Create KMeans model with 3 clusters
kmeans = KMeans(n_clusters=3, random_state=42)

# Fit model
kmeans.fit(X_scaled)

# Get cluster labels
kmeans_labels = kmeans.labels_

# Calculate Silhouette Score
kmeans_score = silhouette_score(X_scaled, kmeans_labels)

print("\nK-Means Silhouette Score:", kmeans_score)

# Visualize K-Means Clusters

plt.figure()
plt.scatter(X_scaled[:, 0], X_scaled[:, 1], c=kmeans_labels)
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.title("K-Means Clustering")
plt.show()

# HIERARCHICAL CLUSTERING

hierarchical = AgglomerativeClustering(n_clusters=3)

hier_labels = hierarchical.fit_predict(X_scaled)

# Calculate Silhouette Score
hier_score = silhouette_score(X_scaled, hier_labels)

print("Hierarchical Clustering Silhouette Score:", hier_score)

# Visualize Hierarchical Clusters

plt.figure()
plt.scatter(X_scaled[:, 0], X_scaled[:, 1], c=hier_labels)
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.title("Hierarchical Clustering")
plt.show()

# Plot Dendrogram
```

```
linked = linkage(X_scaled, method='ward')
```

```
plt.figure(figsize=(10, 5))
dendrogram(linked)
plt.title("Dendrogram (Hierarchical Clustering)")
plt.xlabel("Samples")
plt.ylabel("Distance")
plt.show()
```

```
# Compare Silhouette Scores
```

```
models = ["K-Means", "Hierarchical"]
scores = [kmeans_score, hier_score]
```

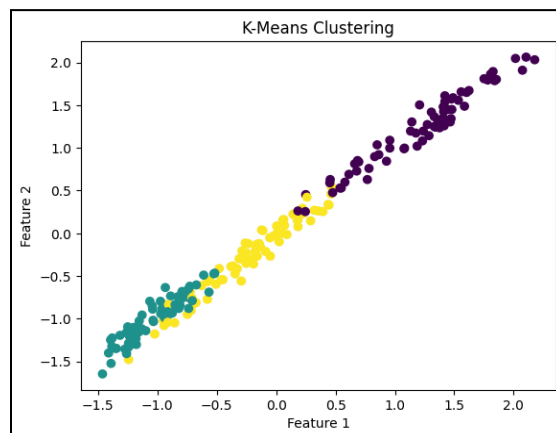
```
plt.figure()
plt.bar(models, scores)
plt.xlabel("Clustering Method")
plt.ylabel("Silhouette Score")
plt.title("Clustering Performance Comparison")
plt.show()
```

Output:

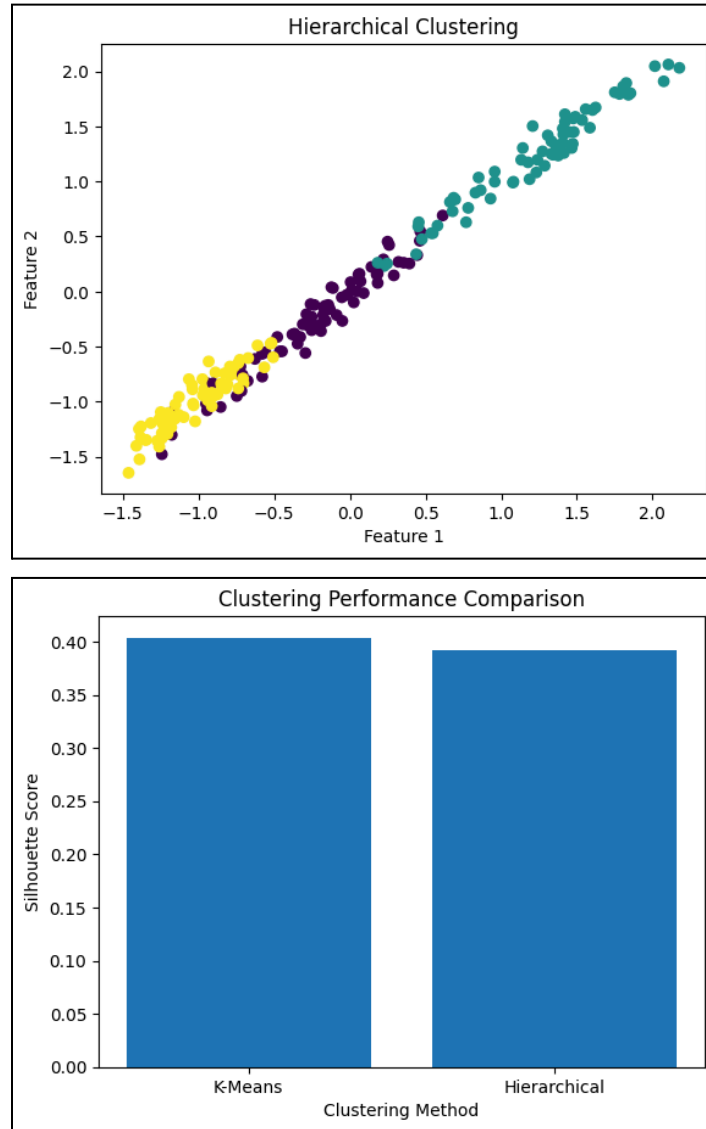
```
Using Colab cache for faster access to the 'seeds-dataset' dataset.
Dataset Path: /kaggle/input/seeds-dataset
Files: ['seeds_dataset.txt']
Dataset Shape: (209, 8)
15.26 14.84 0.871 5.763 3.312 2.221 5.22 1
0 14.68 14.57 0.8811 5.554 3.333 1.018 4.956 1
1 14.29 14.09 0.9050 5.291 3.337 2.699 4.825 1
2 13.84 13.94 0.8955 5.324 3.379 2.259 4.805 1
3 16.14 14.99 0.9034 5.658 3.562 1.355 5.175 1
4 14.38 14.21 0.8951 5.386 3.312 2.462 4.956 1
```

```
Shape of X: (209, 7)
Shape of y: (209,)
Shape of X_scaled: (209, 7)
```

```
K-Means Silhouette Score: 0.403775352942465
```



Hierarchical Clustering Silhouette Score: 0.3922387796755861



Conclusion:

In this experiment, **K-Means** and **Hierarchical Clustering** were applied to the **Seeds dataset** for unsupervised learning. After feature scaling, both methods successfully formed meaningful clusters.

K-Means achieved a **Silhouette Score of 0.4038**, while Hierarchical Clustering achieved **0.3922**, indicating good cluster separation. Overall, K-Means proved computationally efficient, whereas Hierarchical Clustering provided better interpretability through the dendrogram.